

A stylised model of strategic intraday repositioning under capacity, dual-priced imbalance settlement, and the ISP window

Abstract

A two-subperiod, two-market sketch illustrates how three Spanish regime changes — ISP60→ISP15 (2024-12-01), MTU60→MTU15 intraday (2025-03-19) and DA60→DA15 (2025-10-01) — map to the sign and magnitude of the dominant firm’s net intraday position. Two primitives do the work: a physical capacity and the Spanish dual-pricing rule. The imbalance-settlement window (ISP60 vs ISP15) governs whether within-hour imbalances can be netted before settlement, and therefore how strongly the firm prices its structural day-ahead over-commitment into intraday trade.

1 Setup

Two subperiods per hour, $t \in \{1, 2\}$, with physical capacity $K_t > 0$ and marginal cost c . Three stages.

DA. The firm commits a block q_{1t}^{phys} at price p_1 . Under DA60, the block is uniform across subperiods: $q_{11}^{\text{phys}} = q_{12}^{\text{phys}} = q_1/2$. Under DA15, the two commitments are chosen independently.

Intraday. The firm trades q_{2t} at p_{2t} against a linear residual demand with slope b_{2t} : $q_{2t} = b_{2t}(p_1 - p_{2t})$.

Real time. Let $q_t^s \equiv q_{1t}^{\text{phys}} + q_{2t}$ be the firm’s scheduled position. It produces $Q_t \in [0, K_t]$ and generates an imbalance $\Delta_t = Q_t - q_t^s$, which is settled against a random system state $Z \in \{H, L\}$, $\Pr(Z = L) = \pi$, by the Spanish directional rule:

$$p^{\text{imb}} = \begin{cases} p_1 & \text{deviation helpful to the system,} \\ p^{\text{up}} & \text{harmful short } (\Delta < 0, Z = L), \\ p^{\text{dn}} & \text{harmful long } (\Delta > 0, Z = H), \end{cases} \quad p^{\text{dn}} \leq p_1 \leq p^{\text{up}}.$$

Write the reserve spreads as $\phi^+ \equiv p^{\text{up}} - p_1$ and $\phi^- \equiv p_1 - p^{\text{dn}}$. Settlements aggregate over an ISP window $\omega \in \{60, 15\}$: under ISP60 the firm pays against the hourly net $\Delta_H = \Delta_1 + \Delta_2$, so a slack slot cancels a short slot; under ISP15 each Δ_t is settled independently, and no within-hour offset is possible.

2 Effective intraday marginal cost

On the short branch the firm is paid at least p_1 per MWh produced, so it produces up to capacity whenever its schedule q_t^s exceeds K_t . In that *oversell* regime the imbalance is mechanical, $\Delta_t = K_t - q_t^s < 0$, and the expected cost of being short pins down the firm’s effective intraday marginal cost.

Let $\kappa(\omega) \in [0, 1]$ denote the share of the firm's hourly gross short that is *not* cancelled by within-hour offsets:

$$\kappa(\omega) = \begin{cases} \frac{|\Delta_1 + \Delta_2|}{|\Delta_1| + |\Delta_2|} & \omega = 60, \\ 1 & \omega = 15. \end{cases} \quad (1)$$

Under DA60, the block constraint forces $q_{11}^{\text{phys}} = q_{12}^{\text{phys}}$, so whenever capacity varies within the hour ($K_1 \neq K_2$) the firm is short in the tight slot and slack in the loose one. ISP60 lets it net; ISP15 does not.

Proposition 1 (Effective intraday marginal cost). *On the oversell branch,*

$$\text{MC}^*(\omega) = p_1 + \pi \phi^+ \kappa(\omega).$$

On the undersell branch — achieved under DA15 by the slot-by-slot match $q_{1t}^{\text{phys}} = K_t$ —

$$\text{MC}^* = p_1 - (1 - \pi) \phi^-.$$

Proof. On the oversell branch, $\Delta_t = K_t - q_t^s < 0$; producing $Q_t = K_t$ is optimal since $p_1 > c$. Increasing the intraday sale q_2 by one unit raises q^s by one and deepens the short by one. A fraction $\kappa(\omega)$ of this extra short is exposed at settlement; of that fraction the firm pays ϕ^+ with probability π (harmful-short state $Z = L$). The first-order condition for the intraday sale therefore requires

$$p_2 = p_1 + \pi \phi^+ \kappa(\omega) \equiv \text{MC}^*(\omega).$$

On the undersell branch the firm sets $q_{1t}^{\text{phys}} = K_t$ and produces exactly its schedule, so $\Delta_t \geq 0$. A marginal increase in q^s displaces a unit that would otherwise be beneficial-long; it costs $(1 - \pi)\phi^-$ in expectation (harmful-long state $Z = H$), giving $\text{MC}^* = p_1 - (1 - \pi)\phi^-$. $\square$

If $\phi^+ = \phi^- = 0$, then $\text{MC}^* \equiv p_1$ and imbalance settlement drops out.

3 Regime map

In the intraday stage the firm takes p_1 as given and maximises profit against the linear residual demand of each subperiod. For subperiod t the intraday clearing price is $p_{2t} = p_1 - q_{2t}/b_{2t}$, so the first-order condition gives $q_{2t} = \frac{b_{2t}}{2}(p_1 - \text{MC}^*)$. Summing over $t \in \{1, 2\}$ and writing $\beta \equiv b_{22}/b_{21} \in (0, 1]$ for the within-hour residual-demand asymmetry and $\Phi(\beta) \equiv (1 + \beta)/2$:

$$q_{21} + q_{22} = b_{21} \Phi(\beta) (p_1 - \text{MC}^*). \quad (2)$$

The four Spanish regimes map as follows.

Regime	Dates	$\text{MC}^* - p_1$	$\text{sign}(q_{21} + q_{22})$
R1 DA60/ID60, ISP60	pre 2024-12-01	$+\pi\phi^+ \kappa_{60}$	—
R2 DA60/ID60, ISP15	2024-12-01 $\rightarrow$ 2025-03-19	$+\pi\phi^+$	—
R3 DA60/ID15, ISP15	2025-03-19 $\rightarrow$ 2025-10-01	$+\pi\phi^+$	—
R4 DA15/ID15, ISP15	post 2025-10-01	$-(1 - \pi)\phi^-$	+

The sign flip at R4. Under DA15 the firm matches q_{1t}^{phys} to K_t slot-by-slot, closing the structural oversell. It enters the undersell branch and $\text{MC}^* < p_1$, so $q_{21} + q_{22} > 0$. This flip is closed-form: it needs only the mechanical removal of the DA block and the dual-pricing rule, not a calibration of $\phi^\pm$ or π .

The ISP15 channel (R1→R2): predicted but weak in the data. ISP15 sets $\kappa = 1$ and strictly raises the imbalance cost of a given DA overcommitment. In the model, firms internalise this at the DA stage: the equilibrium q_1 under ISP15 is strictly lower than under ISP60, shrinking the structural short and hence $|q_{21} + q_{22}|$ through (2). Empirically, however, this channel barely fires at the 2024-12-01 break. Conditioning on low wind (where the oversell branch should bind most tightly), the OMIE-classified dominant firms’ per-unit intraday repositioning ΔQ is essentially flat across ISP60→ISP15: -268 MWh/unit-day in the DA60/ID60 3-session window (2024-06-14 → 2024-11-30) versus -271 MWh/unit-day in the DA60/ID60 ISP15 window (2024-12-01 → 2025-03-19). The unconditional and firm-level aggregates show the same pattern. One reading consistent with the mechanism but with a second-order calibration: dominants’ DA bids are driven primarily by residual-demand curvature (market-power rents) rather than by the expected imbalance penalty, so a sharper penalty remains second-order as long as firms operate far from the penalty-binding region. The penalty only binds once within-hour repositioning is simultaneously restricted — i.e. jointly with MTU15 (see R2→R3 below).

The thinness channel (R2→R3): the dominant empirical compression. MTU15 in the intraday market introduces asymmetry between the two subperiods’ residual-demand slopes; $\Phi(\beta)$ falls below its $\beta = 1$ value, further compressing $|q_{21} + q_{22}|$ without flipping its sign. This is where the compression visible in the Spanish data actually occurs: at the 2025-03-19 break, low-wind dominant per-unit ΔQ falls sharply from -271 to -78 MWh/unit-day, a 71% compression in a single step. The ordering holds unconditionally, under firm-level aggregation using OMIE’s `grupo.empresarial` labels (both an all-labeled-firms and a Big-4-only definition of dominance), and under a balanced-panel restriction to firms present in every regime, so composition effects and renewable buildout are ruled out as primary drivers. The channel operates on two fronts at once: MTU15 lowers $\Phi(\beta)$ through within-hour asymmetry in residual-demand slopes, and it simultaneously activates the ISP15 penalty by removing the firm’s ability to reposition at a coarser granularity than the settlement grid. What the ISP60→ISP15 transition failed to deliver in isolation (R1→R2 essentially flat), it delivers jointly with intraday granularity at R2→R3.

4 Summary

Two primitives — capacity and dual-priced settlement — plus the ISP window and the DA granularity generate the sign sequence $(-, -, -, +)$ across R1–R4, placing the ISP15 and DA15 reforms on distinct structural channels: ISP15 compresses the magnitude through the DA-side response to a higher effective marginal cost of overselling; DA15 mechanically closes the oversell and flips the sign.

The empirical picture in the Spanish data, using OMIE’s firm classification and per-unit repositioning ΔQ , departs from the model’s structural prose in two respects. First, the ISP15 channel (R1→R2) barely fires in isolation — the compression at 2024-12-01 is essentially nil once wind is held constant — which we read as dominants bidding far from the penalty-binding region, so that a change in κ alone is a second-order shift in their DA programme. Second, the bulk of the compression falls at R2→R3 (MTU15 intraday, 2025-03-19), where the thinness channel and the penalty channel activate jointly: within-hour asymmetry reduces $\Phi(\beta)$ and the firm loses the coarse-granularity hedge that kept the ISP15 penalty slack under DA60/ID60. The thinness channel is therefore the load-bearing empirical channel in Spain, not the ISP-window channel in isolation.

The sign flip predicted at R4 (DA15/ID15, post 2025-10-01) is not observed in the first three months of data: dominant net intraday positions remain negative (Big-4 $\Delta Q \approx -11,500$ MWh/firm-month). This is consistent with either ‘flip delayed’ (transitional frictions — unit-by-unit capacity-matching under DA15 is operationally non-trivial and may take several months of learning) or ‘flip absent’ (the oversell branch remains active for reasons the model does

not capture, e.g. reliability-driven capacity withholding or inherited DA commitment routines). Three months of DA15/ID15 data cannot distinguish the two; the test should be revisited with at least six months of post-2025-10-01 evidence.

Coefficients are directly observable from ESIOS (reserve-activation prices, system imbalance sign) once b_{21} is pinned down from the base model; the empirical reassessment above concerns which channel dominates the Spanish compression, not whether the structural mapping itself survives.

References

- Ito, K. and Reguant, M. (2016). Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–1957.
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A Full model and arbitrage cases

A.1 Firm’s optimisation problem

The dominant firm is a *residual monopolist*: in each market stage it takes the supply of competitive fringe suppliers as given. Fringe suppliers bid at marginal cost, so the dominant firm faces a downward-sloping residual demand.

Day-ahead stage. The firm commits block output q_{1t}^{phys} at the uniform DA clearing price p_1 , which clears the market against fringe supply. Total DA residual demand (summing across subperiods) is $D_1(p_1) = A - b_1 p_1$, where A is total forecasted demand and fringe supply has slope b_1 . Under DA60, the hourly block forces $q_{11}^{\text{phys}} = q_{12}^{\text{phys}} = q_1/2$; under DA15 each subperiod is chosen independently.

Intraday stage. Taking p_1 and q_1^{phys} as given, the firm chooses q_{2t} in each subperiod (negative q_{2t} means buying back). Fringe suppliers adjust along their supply curve, giving residual demand

$$D_{2t}(p_1, p_{2t}) = b_{2t}(p_1 - p_{2t}), \quad (3)$$

with $b_{2t} \leq b_1$, since production is less flexible closer to delivery. Inverting (3) gives

$$p_{2t} = p_1 - \frac{q_{2t}}{b_{2t}}. \quad (4)$$

Combining intraday revenue with the expected imbalance-settlement cost from Proposition 1, the firm’s intraday objective is

$$\max_{q_{21}, q_{22}} \sum_{t=1}^2 [p_{2t} q_{2t} - \text{MC}^*(\omega) q_{2t}]. \quad (5)$$

Substitute (4) into the objective and expand the product:

$$\begin{aligned} \sum_{t=1}^2 [p_{2t} q_{2t} - \text{MC}^* q_{2t}] &= \sum_{t=1}^2 \left[\left(p_1 - \frac{q_{2t}}{b_{2t}} \right) q_{2t} - \text{MC}^* q_{2t} \right] \\ &= \sum_{t=1}^2 \left[(p_1 - \text{MC}^*) q_{2t} - \frac{q_{2t}^2}{b_{2t}} \right]. \end{aligned}$$

The subperiods decouple. Differentiating with respect to q_{2t} and setting the derivative to zero:

$$(p_1 - \text{MC}^*) - \frac{2 q_{2t}}{b_{2t}} = 0 \quad \implies \quad q_{2t} = \frac{b_{2t}}{2} (p_1 - \text{MC}^*(\omega)).$$

Summing the two first-order conditions and collecting:

$$\begin{aligned} q_{21} + q_{22} &= \frac{b_{21}}{2} (p_1 - \text{MC}^*) + \frac{b_{22}}{2} (p_1 - \text{MC}^*) \\ &= \frac{b_{21} + b_{22}}{2} (p_1 - \text{MC}^*) \\ &= b_{21} \underbrace{\frac{1+\beta}{2}}_{\Phi(\beta)} (p_1 - \text{MC}^*), \quad \beta \equiv b_{22}/b_{21}, \end{aligned}$$

which is equation (2) in the main text. A negative net position ($q_{21} + q_{22} < 0$) means the firm repurchases in intraday what it oversold in DA—the Ito and Reguant (2016) *withholding* signature.

Day-ahead stage (backward induction). Anticipating the intraday response, the firm maximises total two-stage profit. Evaluating the intraday equilibrium profit at the optimum: since $p_{2t}^* - \text{MC}^* = (p_1 - \text{MC}^*)/2$ and $q_{2t}^* = (b_{2t}/2)(p_1 - \text{MC}^*)$,

$$\sum_t (p_{2t}^* - \text{MC}^*) q_{2t}^* = \sum_t \frac{b_{2t}}{4} (p_1 - \text{MC}^*)^2 = \frac{b_{21} + b_{22}}{4} (p_1 - \text{MC}^*)^2.$$

Writing $p_1(q_1) = (A - q_1)/b_1$ (inverted DA residual demand), the firm's reduced-form programme is

$$\max_{q_1} \Pi(q_1) = (p_1(q_1) - c) q_1 + \frac{b_{21} + b_{22}}{4} (p_1(q_1) - \text{MC}^*)^2.$$

Using $dp_1/dq_1 = -1/b_1$, the first-order condition is

$$\underbrace{(p_1 - c) - q_1/b_1}_{\text{DA monopoly margin}} - \underbrace{\frac{b_{21} + b_{22}}{2b_1} (p_1 - \text{MC}^*)}_{\text{intraday-profit response}} = 0.$$

With linear residual demand and constant marginal cost c , the equilibrium satisfies $\text{MC}^* > p_1 > c$ on the oversell branch (the analogue of Result 1 in Ito and Reguant 2016): the firm exercises DA markup over c , and the imbalance-settlement premium $\text{MC}^* - p_1 = \pi \phi^+ \kappa(\omega) > 0$ discourages further IDA sales so $q_{2t} < 0$. Substituting $\text{MC}^* = p_1 + \pi \phi^+ \kappa(\omega)$ into equation (2):

$$\begin{aligned} p_1 - \text{MC}^* &= p_1 - [p_1 + \pi \phi^+ \kappa(\omega)] = -\pi \phi^+ \kappa(\omega), \\ q_{21} + q_{22} &= b_{21} \Phi(\beta) \cdot (-\pi \phi^+ \kappa(\omega)) = -b_{21} \Phi(\beta) \pi \phi^+ \kappa(\omega) < 0, \end{aligned}$$

so the dominant firm is a systematic net buyer in intraday.

A.2 Fringe arbitrage and capacity constraints

Following Ito and Reguant (2016), fringe suppliers can exploit the DA premium by overselling quantity s in DA and buying it back in intraday. An arbitrageur shifting s modifies residual demands to

$$D_1(p_1, s) = A - b_1 p_1 - s, \quad D_{2t}(p_1, p_{2t}, s) = b_{2t}(p_1 - p_{2t}) + s/2,$$

with $s/2$ split equally across subperiods. Inverting gives

$$p_1(q_1, s) = \frac{A - q_1 - s}{b_1}, \quad p_{2t}(q_{2t}, s) = p_1 - \frac{q_{2t} - s/2}{b_{2t}}.$$

Case 1: No arbitrage ($s = 0$). The dominant firm exploits the full DA premium. Evaluating equation (2) on the oversell branch:

$$q_{21} + q_{22} = -b_{21} \Phi(\beta) \pi \phi^+ \kappa(\omega),$$

which is the maximum-magnitude net position in a given regime.

Case 2: Full competitive arbitrage. With many fringe wind farms, s rises until the no-arbitrage condition $p_1 = p_{2t}$ holds for each t . Setting $p_{2t} = p_1$ in the inverse demand gives

$$0 = -\frac{q_{2t} - s/2}{b_{2t}} \implies q_{2t}^{\text{arb}} = s/2,$$

so the arbitrageur's buy-back in each subperiod exactly matches the dominant firm's IDA sale. The dominant firm's IDA FOC $q_{2t} = (b_{2t}/2)(p_1 - \text{MC}^*)$ is unchanged, but when the no-arbitrage condition binds the effective residual demand it faces in IDA has slope $(b_1 + b_{21} + b_{22})$ across the combined market. At $p_1 = p_2$, the firm is indifferent across stages and chooses total output $q_1 + q_{21} + q_{22}$ to solve a single monopoly problem against the pooled slope. Substituting into the firm's DA FOC yields a lower equilibrium q_1 — the firm *withholds more* in DA — and a net intraday position

$$q_{21} + q_{22} = \frac{b_{21} + b_{22}}{b_1 + b_{21} + b_{22}} \cdot (\text{pooled markup adjustment}),$$

compressed relative to Case 1 but still negative so long as $\text{MC}^* > p_1$. This is the mechanism behind Result 2 of Ito and Reguant (2016): full arbitrage converges prices, forces the dominant firm to withhold more in DA, and shrinks total output relative to the no-arbitrage benchmark.

Case 3: Limited arbitrage ($s \leq K$). In practice, arbitrage is constrained by:

- *Physical capacity*: fringe firms cannot sell beyond installed capacity in DA.
- *Regulatory limits*: OMIE discourages systematic large swings in scheduled production unless justified by technical operations (Ito and Reguant, 2016).
- *MTU granularity*: under MTU15, there are four delivery subperiods per hour instead of one, multiplying the number of (s, π) slots the arbitrageur must manage and potentially increasing coordination costs.

When $s = K$ binds, the arbitrageur's residual-demand shift is capped. Taking the DA-FOC and IDA-FOC at $s = K$, the DA-IDA wedge solves

$$p_1 - p_{2t} = \frac{q_{2t} - K/2}{b_{2t}},$$

so for any $q_{2t} < K/2$ (the dominant firm under-arbitrages relative to the fringe capacity) the wedge is strictly positive. Following Result 3 of Ito and Reguant (2016), substituting the dominant firm's FOC and solving for the equilibrium residual premium gives comparative statics:

$$\frac{\partial(p_1 - p_2)}{\partial A} > 0, \quad \frac{\partial(p_1 - p_2)}{\partial b_1} < 0, \quad \frac{\partial(p_1 - p_2)}{\partial b_{2t}} < 0,$$

i.e. DA premia persist and are larger when forecasted demand A is high, fringe supply is inelastic (low b_1), or the intraday market is elastic (high b_{2t}). All three comparative statics carry directly into the reform narrative.

A.3 Regime map and the four channels

The four Spanish regimes map onto the Ito–Reguant framework as follows.

- R1 (DA60/ID60, ISP60).** Within-hour netting ($\kappa_{60} < 1$) lowers the imbalance cost of a given DA overcommitment, so MC^* is close to p_1 and the withholding incentive is moderated. Fringe wind farms arbitrage the DA premium; the arbitrage capacity constraint K is typically binding, leaving a residual premium. Net intraday position: $q_{21} + q_{22} < 0$.
- R2 (DA60/ID60, ISP15).** ISP15 sets $\kappa = 1$, strictly raising $MC^* = p_1 + \pi\phi^+$. The dominant firm internalises the higher cost at the DA stage and reduces q_1 , compressing $|q_{21} + q_{22}|$ through equation (2) without a sign flip. Fringe arbitrage capacity is unchanged, but it operates against a smaller residual premium. Net position: $q_{21} + q_{22} < 0$, smaller in magnitude.
- R3 (DA60/ID15, ISP15).** MTU15 intraday introduces within-hour asymmetry: the second subperiod’s residual demand slope falls ($b_{22} < b_{21}$, i.e. $\beta < 1$), lowering $\Phi(\beta)$ and further compressing $|q_{21} + q_{22}|$ without changing its sign. The dominant firm does not change branches; it simply faces a thinner intraday market for repositioning. Net position: $q_{21} + q_{22} < 0$, further compressed.
- R4 (DA15/ID15, ISP15).** DA15 allows the firm to set $q_{1t}^{\text{phys}} = K_t$ slot-by-slot, closing the structural oversell. The firm moves to the undersell branch: $MC^* = p_1 - (1 - \pi)\phi^- < p_1$, so $p_1 - MC^* > 0$ and $q_{21} + q_{22} > 0$. The dominant firm becomes a *net seller* in intraday, and fringe firms that previously arbitrated by buying back in IDA now face the opposite premium structure. Net position: $q_{21} + q_{22} > 0$ (sign flip).

The regime table in Section 3 summarises these four cases. The sign sequence $(-, -, -, +)$ is generated entirely by the two primitives (physical capacity and dual-priced settlement) interacting with three institutional changes; it does not require calibration of $\phi^\pm$, π , or b_{2t} .